FlipValue

Independent phone resale intelligence for India, with the US as a second market



Project overview · v0.2.0 · October 2026

Five questions, one honest answer each

1	**What is my phone worth?**	As-is private-sale value from comparable sales
2	**What would a dealer pay?**	Buyback and trade-in quotes, kept separate
3	**What if I repair it?**	After-repair value, only with real evidence
4	**What does it cost new or refurbished?**	Cross-source price comparison
5	**Repair or sell?**	Repair-vs-sell arithmetic in one calculator

One FastAPI service, four front doors

Website

`/`

React app: value a phone, compare prices, trends, watchlist

Developer API

`/api/v1`

X-API-Key with scopes, per-key limits and daily quota

Partner API

`/v1/valuations`

FixMyTech contract, idempotent, optional client

Operator API

`/admin/api`

Sources, imports, rejects, audit, offer refresh

▼

`engine.py` one valuation engine serves every channel; the frontend holds no pricing logic

Valuation method: comparables-v1

1

Select

Same model, storage, region and grade within a 180-day window. Nothing pooled.

2

Pick a basis

Completed sales, then listings, then synthetic (dev only). Bases never mixed.

3

Summarise

Median value. n ≥ 10: 10th–90th percentile; 3–9: min–max; under 3: no estimate.

4

Label quality

Evidence-backed needs completed sales, n ≥ 10, newest ≤ 90 days. Else limited.

Faults and device age do not adjust prices. There is no evidence for fixed discounts, and every result says so.

Value types are never blended

Private sale

Median of comparable resale observations

Buyback

Dealer offers, then trade-in, summarised separately

Refurbished retail

What retailers charge now, from marketplace offers

Original retail

MRP, for context only

After repair

Uplift from comparables; unavailable without post-repair evidence

Each result carries confidence class, assumptions, limitations and how to improve it.

Marketplace data and product matching

Source	Status
Flipkart Affiliate API	Adapter tested on fixtures; blocked live (needs approved account)
Amazon.in Creators API	Adapter tested on fixtures; blocked live (needs Associates approval)
Cashify	Manual quotes only; terms prohibit systematic retrieval
Real completed sales	Blocked: no authorised source yet

Match categories

Exact

Strong

Variant

Similar

Unmatched

Every match carries a score and reasons you can inspect.

Developer platform and API keys

Keys and access

- Accounts: email plus scrypt, HttpOnly session cookie, CSRF header
- Keys from a CSPRNG, shown once, stored as SHA-256
- Scopes enforced server-side; revoke takes effect on the next request
- No raw key ever appears in the audit log

/api/v1 endpoints

- health
- valuations (create, get; owner-only)
- market-prices and price-history
- products
- usage, with per-key minute limits, daily quotas and X-RateLimit headers

Where it stands today

222
backend tests passing

8
skipped (no PostgreSQL server)

6
catalogue models, hand-curated

2
migrations (0002 not run on PostgreSQL)

Also covered: Vitest and Testing Library, Playwright end-to-end in Edge, ruff lint, typecheck. Not production-ready and not deployed.

Known limits, stated up front

No real market data

Every number in dev is synthetic and labelled Mock; production returns no data until a source is imported.

Small catalogue

6 models; unknown OEM variants return 404 until an alias is added.

Self-reported condition

The questionnaire maps answers to a grade; nothing verifies hardware.

Account gaps

No email verification, password reset or self-service deletion.

Not deployed

Cloud config rehearsed locally; image build, services and HTTPS unverified.

Drafts

Privacy and Terms pages have had no legal review.

What unlocks production

1. Secure approved Flipkart and Amazon accounts
2. Import an authorised source of completed sales
3. Run migration 0002 on PostgreSQL
4. Deploy to Firebase, Cloud Run and Cloud SQL
5. Add email verification and legal review of Terms